

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech (ME) Examination

Nov – Dec 2018

ME416 Operations Research (Elective-I)

Date: 30.11.2018, Friday

Time: 01.30 p.m. to 04.30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION –I**Q-1 Answer the following multiple choice questions. (Each carries one mark) [07]**

- (1) To formulate a problem for solution by the simplex method, we must add artificial variable to-
(a) Only equality constraints (b) Only 'greater than' constraints (c) Both (a) and (b)
- (2) If two constraints do not intersect in the positive quadrant of the graph, then-
(a) The solution is infeasible (b) The solution is unbounded
(c) One of the constraints is redundant (d) None of these
- (3) The solution region of an LP problem is unbounded due to-
(a) An incorrect formulation of the LP problem (b) Objective function is unbounded
(c) Neither (a) nor (b) (d) Both (a) and (b)
- (4) For maximization LP model, the simplex method is terminated when all values-
(a) $C_i - Z_j \leq 0$ (b) $C_i - Z_j \geq 0$ (c) $C_i - Z_j = 0$ (d) $Z_j \leq 0$
- (5) Customer behavior in which the customer moves from one queue to another in a multiple channel situation is-
(a) Balking (b) Reneging (c) Jockeying (d) Alternating
- (6) _____ is the mathematical technique of optimizing a sequence of interrelated decisions over a period of time. (fill in the blank)
(a) Linear programming (b) Dynamic programming (c) Queuing (d) PERT
- (7) A stage in a dynamic programming problem represents-
(a) Number of decision alternatives (b) Different time periods in the planning
(c) Status of the system at a particular state (d) All of these

Q-2 (a) Maximize $Z = X_1 + 2X_2 + 3X_3 - X_4$ [06]
Subject to the constraints:

$$X_1 + 2X_2 + 3X_3 = 15, \quad 2X_1 + X_2 + 5X_3 = 20, \quad X_1 + 2X_2 + X_3 + X_4 = 10, \\ X_1, X_2, X_3, X_4 \geq 0.$$

OR**Q-2 (a) Solve the following LP problem by using simplex method. [06]**

Minimize $Z = 2Y_1 + 3Y_2$

Subject to the constraints:

$$Y_1 + Y_2 \geq 5, \quad Y_1 + 2Y_2 \geq 6, \quad Y_1, Y_2 \geq 0.$$

Q-2 (b) Formulate following problem and solve by simplex method- [08]

ABC printing company is facing a tight financial squeeze and is attempting to cut costs wherever possible. At present it has only one printing contract, and luckily, the book is selling well in both the hardcover and paperback editions. It has just received a request to print more copies of this book in either the hardcover or the paperback form. The printing cost for the hardcover books is Rs. 600 per 100 books while that for paperback is only Rs. 500 per 100 books. Although the company is attempting to economize, it

does not wish to lay off any employee. Therefore, it feels obliged to run its two printing presses- I and II, at least 80 and 60 hours per week, respectively. Press I can produce 100 hardcover books in 2 hours or 100 paperback books in 1 hour. Press II can produce 100 hardcover books in 1 hour or 100 paperbacks books in 2 hours. Determine how many books of each type should be printed in order to minimize costs.

- Q-3 (a)** Write down the various applications of OR (Operations Research) in industry. [03]
- Q-3 (b)** D-Mart employs one cashier at its counter. 8 customers arrive on an average every 5 mins, while the cashier can serve 10 customers in the same time. Assuming Poisson distribution for arrival and exponential distribution for service rate, determine- [05]
- (1) Average numbers of customers in the system.
 - (2) Average numbers of customers in queue.
 - (3) Average time a customer spends in the system.
 - (4) Average time a customer waits before being served.
- Q-3 (c)** Use dynamic programming to solve the following linear programming problem- [06]
- Maximize, $Z = 3X_1 + 4X_2$
 Subject to the constraints: $2X_1 + 3X_2 \leq 24$, $X_1 \leq 6$, $X_1, X_2 \geq 0$.
- OR**
- Q-3 (c)** Use dynamic programming to solve the following linear programming problem- [06]
- Maximize, $Z = 2X_1 + 5X_2$
 Subject to the constraints: $2X_1 + X_2 \leq 430$, $2X_2 \leq 460$, $X_1, X_2 \geq 0$.

SECTION –II

- Q-4 Answer the following multiple choice questions. (Each carries one mark)** [07]
- (1) Which of the following methods is used to verify the optimality of the current solution of the transportation problem?
 - (a) Least cost method
 - (b) Vogel's approximation method
 - (c) Modified distribution method
 - (d) All of these
 - (2) The method used for solving an assignment problem is called-
 - (a) Reduced matrix method
 - (b) Modified distribution method
 - (c) Hungarian method
 - (d) None of these
 - (3) An optimal solution of an assignment problem can be obtained only if-
 - (a) Each row and column has only one zero element
 - (b) Each row and column has at least one zero element
 - (c) The data arrangement is in a square matrix
 - (d) None of these
 - (4) If the maximin value is same as the minimax value, then the game is said to have a saddle point.
 - (a) True
 - (b) False
 - (5) The size of the payoff matrix of a game can be reduced by using the principle of-
 - (a) Game inversion
 - (b) Rotation reduction
 - (c) Dominance
 - (d) Game transpose
 - (6) If an activity has zero slack, it implies that-
 - (a) It lies on the critical path
 - (b) It lies on the non-critical path
 - (c) The project is progressing well
 - (d) None of these
 - (7) Generally the PERT technique deals with the project of-
 - (a) Repetitive nature
 - (b) Non- repetitive nature
 - (c) Deterministic nature
 - (d) None of these

- Q-5 (a)** Four lecturers, each capable of teaching any one of the four different subjects. Class preparation time in hours for different subjects varies from teacher to teacher and is- [06]

given in the table below. Each lecturer is assigned only one subject. Determine an assignment schedule so as to minimize the total subject preparation time for all subjects.

Lecturer	Thermo-Dynamics	Machine Design	Production Engineering	Operations Research
A	2	10	9	7
B	15	4	14	8
C	13	14	16	11
D	4	15	13	9

- Q-5 (b)** The utility data for a network are given below. Determine the total float and identify the critical path. [06]

Activity	0-1	1-2	1-3	2-4	2-5	3-4	3-6	4-7	5-7	6-7
Duration (hours)	2	8	10	6	3	3	7	5	2	8

OR

- Q-5 (b)** A project schedule has the following characteristics- [06]

Activity	Time (weeks)	Activity	Time (weeks)
1-2	4	5-6	4
1-3	1	5-7	8
2-4	1	6-8	1
3-4	1	7-8	2
3-5	6	8-10	5
4-9	5	9-10	7

Construct the network and find critical path with project duration.

- Q-6 Answer following questions. (Any two)** [16]

- (a) Find the optimum solution to the following transportation problem in which the cells contain the transportation cost in rupees.

	W_1	W_2	W_3	W_4	W_5	Available
F_1	7	6	4	5	9	40
F_2	8	5	6	7	8	30
F_3	6	8	9	6	5	20
F_4	5	7	7	8	6	10
Required	30	30	15	20	5	

- (b) Reduce the following game and find the game value-

		Player B			
		I	II	III	IV
Player A	I	3	2	4	0
	II	3	4	2	4
	III	4	2	4	0
	IV	0	4	0	8

- (c) The maintenance cost and resale value per year of a machine whose purchase price is Rs. 7000 is given below-

Year	1	2	3	4	5	6	7	8
Maintenance cost in Rs.	900	1200	1600	2100	2800	3700	4700	5900
Resale value in Rs.	4000	2000	1200	600	500	400	400	400

When should the machine be replaced?
